

factor

$$\mathcal{A}(q) = \int \frac{d^3p}{(2\pi)^3} a(\mathbf{p}, \mathbf{q}, \omega) L(\mathbf{p}, \mathbf{q}, \omega), \quad (\text{A1})$$

$$\mathcal{B}(q) = \Delta\omega \int \frac{d^3p}{(2\pi)^3} \frac{1}{\epsilon_p} L(\mathbf{p}, \mathbf{q}, \omega), \quad (\text{A2})$$

$$\begin{aligned} \mathcal{C}(q) = & - \int \frac{d^3p}{(2\pi)^3} c(\mathbf{p}, \mathbf{q}, \omega) L(\mathbf{p}, \mathbf{q}, \omega) \\ & - \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\epsilon_p} [1 - 2f(\epsilon_p)], \end{aligned} \quad (\text{A3})$$

where

$$a(\mathbf{p}, \mathbf{q}, \omega) = \frac{\epsilon_p + \epsilon_{p+q}}{2} \left(1 - \frac{\xi_p \xi_{p+q}}{\epsilon_p \epsilon_{p+q}} + \frac{\Delta^2}{\epsilon_p \epsilon_{p+q}} \right), \quad (\text{A4})$$

$$c(\mathbf{p}, \mathbf{q}, \omega) = \frac{\epsilon_p + \epsilon_{p+q}}{2} \left(1 + \frac{\xi_p \xi_{p+q}}{\epsilon_p \epsilon_{p+q}} + \frac{\Delta^2}{\epsilon_p \epsilon_{p+q}} \right), \quad (\text{A5})$$

$$L(\mathbf{p}, \mathbf{q}, \omega) = \frac{1 - f(\epsilon_p) - f(\epsilon_{p+q})}{(\omega + i\delta)^2 - (\epsilon_p + \epsilon_{p+q})^2}. \quad (\text{A6})$$

Next we expand these functions in powers of small parameter y to order $O(y^2)$

$$\begin{aligned} a(\mathbf{p}, \mathbf{q}, \omega) = & \frac{2\Delta^2}{\epsilon_p} - \frac{2\Delta^2 \mu \xi_p}{\epsilon_p^3} x y \\ & + \frac{\Delta^2 \mu \xi_p}{\epsilon_p^5} (6\mu \xi_p x^2 - \epsilon_p^2) y^2, \end{aligned} \quad (\text{A7})$$

$$c(\mathbf{p}, \mathbf{q}, \omega) = 2\epsilon_p + \frac{2\mu \xi_p}{\epsilon_p} x y + \frac{\mu \xi_p}{\epsilon_p} y^2, \quad (\text{A8})$$

$$L(\mathbf{p}, \mathbf{q}, \omega) = L_0 + L_1 x y + L_2 y^2, \quad (\text{A9})$$

where the coefficients of the expansion (A9) can be written using the shorthand expressions $D \equiv \omega^2 - 4\epsilon_p^2$ and $\zeta \equiv \epsilon_p/2T$ as

$$L_0 = \frac{\tanh \zeta}{D + i\delta}, \quad (\text{A10})$$

$$L_1 = 8\mu \xi_p \frac{\tanh \zeta}{(D + i\delta)^2} + \frac{\mu \xi_p}{2\epsilon_p T} \frac{\text{sech}^2 \zeta}{(D + i\delta)}, \quad (\text{A11})$$

$$\begin{aligned} L_2 = & \frac{4\mu^2 x^2 \xi_p^2 \text{sech}^2 \zeta}{\epsilon_p T (D + i\delta)^2} + \frac{\mu^2 x^2 \text{sech}^2 \zeta}{2\epsilon_p T (D + i\delta)} \\ & + \frac{\mu \xi_p \text{sech}^2 \zeta}{4\epsilon_p T (D + i\delta)} - \frac{\mu^2 x^2 \xi_p^2 \text{sech}^2 \zeta}{2\epsilon_p^3 T (D + i\delta)} \\ & + \frac{64\mu^2 x^2 \xi_p^2 \tanh \zeta}{(D + i\delta)^3} + \frac{8\mu^2 x^2 \tanh \zeta}{(D + i\delta)^2} \\ & + \frac{4\mu \xi_p \tanh \zeta}{(D + i\delta)^2} - \frac{4\mu^2 x^2 \xi_p^2 \tanh \zeta}{\epsilon_p^2 (D + i\delta)^2} \\ & - \frac{\mu^2 x^2 \xi_p^2 \tanh \zeta \text{sech}^2 \zeta}{2\epsilon_p^2 T^2 (D + i\delta)}, \end{aligned} \quad (\text{A12})$$

where $\text{sech}^2(x) \equiv 1 - \tanh(x)^2$. On substituting Eqs. (A7) and (A9) in Eq. (A1) we obtain

$$\begin{aligned} \mathcal{A}(q) = & \int \frac{d^3p}{(2\pi)^3} \left[\frac{2\Delta^2}{\epsilon_p} L_0 + \frac{2\Delta^2}{\epsilon_p} L_2 y^2 \right. \\ & \left. - \frac{2\Delta^2 \mu \xi_p}{\epsilon_p^3} L_1 x^2 y^2 + \frac{\Delta^2 \mu \xi_p}{\epsilon_p^5} (6\mu \xi_p x^2 - \epsilon_p^2) L_0 y^2 \right] \\ = & \mathcal{A}_0(q) + \mathcal{A}_2(q) y^2 + O(y^4). \end{aligned} \quad (\text{A13})$$

The terms that are odd in x drop out on integration in symmetrical limits. Similarly,

$$\begin{aligned} \mathcal{C}(q) = & - \int \frac{d^3p}{(2\pi)^3} \left[2\epsilon_p L_0 + \frac{2\mu \xi_p}{\epsilon_p} L_1 x^2 y^2 + \frac{\mu \xi_p}{\epsilon_p} L_0 y^2 \right. \\ & \left. + 2\epsilon_p L_2 y^2 \right] - \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\epsilon_p} [1 - 2f(\epsilon_p)]. \end{aligned} \quad (\text{A14})$$

We further use the relation

$$\begin{aligned} & - \int \frac{d^3p}{(2\pi)^3} 2\epsilon_p L_0 - \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\epsilon_p} [1 - 2f(\epsilon_p)] \\ = & - \frac{\omega^2}{2} \int \frac{d^3p}{(2\pi)^3} \frac{1}{\epsilon_p} L_0 \end{aligned} \quad (\text{A15})$$

to write

$$\begin{aligned} \mathcal{C}(q) = & - \frac{\omega^2}{2} \int \frac{d^3p}{(2\pi)^3} \frac{1}{\epsilon_p} L_0 - \int \frac{d^3p}{(2\pi)^3} \left[\frac{2\mu \xi_p}{\epsilon_p} L_1 x^2 y^2 \right. \\ & \left. + \frac{\mu \xi_p}{\epsilon_p} L_0 y^2 + 2\epsilon_p L_2 y^2 \right] = \mathcal{C}_0(q) + \mathcal{C}_2(q) y^2. \end{aligned} \quad (\text{A16})$$

Finally,

$$\mathcal{B}(q) = \Delta\omega \int \frac{d^3p}{(2\pi)^3} \frac{1}{\epsilon_p} (L_0 + L_2 y^2) = \mathcal{B}_0(q) + \mathcal{B}_2(q) y^2, \quad (\text{A17})$$

and $\mathcal{A}_0/\mathcal{C}_0 = -4\Delta^2/\omega^2$, and $\mathcal{B}_0/\mathcal{C}_0 = -2\Delta/\omega$.

Appendix B: Evaluating phase-space integrals

Here we evaluate the integrals (39)–(41), which are given as

$$I_0 = \nu(p_F)^{-1} \mu \int \frac{d^3p}{(2\pi)^3} \frac{\xi_p}{\epsilon_p^3} \left[\frac{4\epsilon_p^2}{\omega^2} - 1 + \frac{6\mu \xi_p}{\epsilon_p^2} x^2 \right] L_0, \quad (\text{B1})$$

$$I_1 = 2\nu(p_F)^{-1} \mu \int \frac{d^3p}{(2\pi)^3} \frac{\xi_p}{\epsilon_p^3} \left[\frac{4\epsilon_p^2}{\omega^2} - 1 \right] L_1 x^2, \quad (\text{B2})$$

$$I_2 = 2\nu(p_F)^{-1} \mu \int \frac{d^3p}{(2\pi)^3} \frac{1}{\epsilon_p} \left[\frac{4\epsilon_p^2}{\omega^2} - 1 \right] L_2(x). \quad (\text{B3})$$